

Task 1 – Machine Learning Algorithms

Artificial Intelligence Internship – 6-Week AI Internship

This report explains four Machine Learning algorithms: Logistic Regression, Decision Tree Classifier, K-Nearest Neighbors (KNN), and Linear Regression.

1. Logistic Regression

What is it?

Logistic Regression is a supervised Machine Learning classification algorithm used to predict categorical outcomes. For binary classification, it estimates the probability that an observation belongs to one of two classes.

How does it work?

The algorithm calculates a weighted combination of input features and passes the result through a logistic function to produce a probability. A classification threshold is then used to determine the predicted class.

Where is it used?

It is commonly used for binary classification problems in areas such as healthcare, finance, marketing, and risk analysis.

One Real-World Application

A real-world application is medical diagnosis, such as predicting whether a tumor is malignant or benign.

Advantages

- Simple and easy to understand
- Fast to train
- Works well for linearly separable data
- Produces probability estimates

Limitations

- Assumes a relatively linear relationship between features and the log-odds
- May perform poorly on complex non-linear relationships
- Can be affected by irrelevant or highly correlated features

2. Decision Tree Classifier

What is it?

A Decision Tree Classifier is a supervised Machine Learning algorithm used for classification. It represents decisions using a tree-like structure.

How does it work?

The algorithm repeatedly divides the dataset based on feature values. Each internal node represents a decision, branches represent outcomes, and leaf nodes represent predicted classes.

Where is it used?

Decision Trees are used in classification tasks where understandable decision rules are useful.

One Real-World Application

A Decision Tree can be used for medical classification, customer segmentation, loan approval, and fraud detection.

Advantages

- Easy to interpret
- Can model non-linear relationships
- Requires little data preprocessing
- Can provide feature importance

Limitations

- Can overfit the training data
- Small changes in data can produce different trees
- Very deep trees can become difficult to interpret

3. K-Nearest Neighbors (KNN)

What is it?

K-Nearest Neighbors is a supervised Machine Learning algorithm that predicts the class of a new observation based on the classes of nearby observations.

How does it work?

KNN calculates the distance between a new data point and existing training points. It selects the K closest observations and uses their majority class to make the prediction.

Where is it used?

KNN is used for classification and pattern recognition problems.

One Real-World Application

KNN can be used for image recognition, recommendation systems, and identifying similar customer or medical records.

Advantages

- Simple to understand
- Easy to implement
- No complex training process
- Can model non-linear decision boundaries

Limitations

- Prediction can be computationally expensive for large datasets
- Sensitive to feature scaling
- Choosing the appropriate value of K can be difficult
- Sensitive to irrelevant features

4. Linear Regression

What is it?

Linear Regression is a supervised Machine Learning algorithm used to predict continuous numerical values.

How does it work?

The algorithm finds the best-fitting line between input features and a continuous target variable by minimizing the difference between actual and predicted values.

Where is it used?

Linear Regression is commonly used for prediction and forecasting problems involving numerical targets.

One Real-World Application

It can be used to predict student marks based on hours studied, as demonstrated in Task 3 of this internship assignment.

Advantages

- Simple and easy to interpret
- Fast to train
- Works well when relationships are approximately linear
- Provides understandable coefficients

Limitations

- Assumes a linear relationship
- Sensitive to outliers
- May perform poorly with complex relationships
- Can be affected by multicollinearity between features

Algorithm Summary

Algorithm	Type	Main Purpose
Logistic Regression	Classification	Predict categorical classes
Decision Tree Classifier	Classification	Classify observations using decision rules
KNN	Classification	Classify based on nearby observations
Linear Regression	Regression	Predict continuous numerical values

Conclusion: These four algorithms demonstrate important Machine Learning approaches for both classification and regression. Logistic Regression and Decision Tree Classifier are useful for categorical prediction, KNN provides a similarity-based approach, and Linear Regression is suitable for predicting continuous numerical outcomes.